

Business Problem Statement

The e-commerce retailer operates across 20 Indian cities with data spanning 2024 to 2025. Despite broad market coverage, the business lacks clear insights into its primary growth drivers. Management needs to identify which specific regions and cities contribute most to net profitability. A critical challenge is determining if aggressive discounting is hurting or helping the bottom line. The analysis must evaluate the performance of 10 product categories, including Books and Groceries. Inventory management needs optimization based on high-demand sub-categories and regional trends. Customer payment preferences across UPI, Cards, and COD must be analyzed for transaction efficiency. Seasonal fluctuations in sales and profit require a detailed month-over-month investigation. The objective is to maximize the average order value while minimizing loss-making transactions. Ultimately, the company seeks data-driven strategies for regional marketing and optimized pricing.

Deliverables

- 1. Data Preparation & Modeling (Python):** Clean and transform the raw dataset for analysis.
- 2. Data Analysis (SQL):** Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.
- 3. Visualization & Insights (Power BI):** Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.
- 4. Report and Presentation:** Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.
- 5. GitHub Repository:** Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.